

Calculus I

Limits

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Outline

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 - One-sided Limits
- 2 Calculating Limits Using Limit Laws

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The Limit of a Function

Definition (The Limit of a Function)

We write

$$\lim_{x \rightarrow a} f(x) = L$$

and say “the limit of $f(x)$, as x approaches a , equals L ,” if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a (on either side of a) but not equal to a .

Equivalent formulation. $\lim_{x \rightarrow a} f(x) = L$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for all x with $0 < |x - a| < \delta$.

Example

- Guess the value of $\lim_{x \rightarrow 1} \frac{x-1}{x^2-1}$.
- Notice that $\frac{x-1}{x^2-1}$ is not defined at 1.
- It is defined for values of x near 1.
- We guess that the limit is 0.5.
- In this case, our guess is correct.

x	$f(x)$	x	$f(x)$
0.5	0.666667	1.5	0.400000
0.9	0.526316	1.1	0.476190
0.99	0.502513	1.01	0.497512
0.999	0.500250	1.001	0.499750
0.9999	0.500025	1.0001	0.499975

Example

- Guess the value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.
- Notice that $\frac{\sin x}{x}$ is not defined at 0.
- It is defined for all other values of x near 0.
- We guess that the limit is 1.
- In this case, our guess is correct.

x	$f(x)$	x	$f(x)$
± 1.0	0.841471	± 0.1	0.998334
± 0.5	0.958851	± 0.05	0.999583
± 0.4	0.973546	± 0.01	0.999983
± 0.3	0.985067	± 0.005	0.999995
± 0.2	0.993347	± 0.001	0.999999

Example

- Guess the value of $\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right)$.
- Notice that $\sin\left(\frac{\pi}{x}\right)$ is not defined at 0.
- It is defined for values of x near 0.
- We may guess that the limit is 0.
- Such a guess would be **wrong**.

x	$f(x)$	x	$f(x)$
1	$\sin \pi = 0$	$\frac{1}{2}$	$\sin(2\pi) = 0$
$\frac{1}{3}$	$\sin(3\pi) = 0$	$\frac{1}{4}$	$\sin(4\pi) = 0$
0.1	$\sin(10\pi) = 0$	0.01	$\sin(100\pi) = 0$

One-sided Limits

Example

The Heaviside function H is defined by

$$H(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}.$$

- As x approaches 0 from the left, $H(x)$ approaches 0.
- As x approaches 0 from the right, $H(x)$ approaches 1.
- There is no single number that $H(x)$ approaches as x approaches 0.
- Therefore $\lim_{x \rightarrow 0} H(x)$ doesn't exist.

Definition (Left-hand Limit)

We write

$$\lim_{x \rightarrow a^-} f(x) = L \quad \text{or} \quad \lim_{\substack{x \rightarrow a \\ x < a}} f(x) = L$$

and say the left-hand limit of $f(x)$ as x approaches a is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to and less than a .

Definition (Right-hand Limit)

We write

$$\lim_{x \rightarrow a^+} f(x) = L \quad \text{or} \quad \lim_{\substack{x \rightarrow a \\ x > a}} f(x) = L$$

and say the **right**-hand limit of $f(x)$ as x approaches a is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to and **greater** than a .

We can define a **right**-hand limit similarly.

By comparing definitions, we can see that

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L.$$

Example

The graph of a function g is shown to the right. Use it to state the values (if they exist) of the following:

$$\begin{array}{l|l} \lim_{x \rightarrow 1^-} g(x) = 3 & \lim_{x \rightarrow 3^-} g(x) = 1 \\ \lim_{x \rightarrow 1^+} g(x) = 3 & \lim_{x \rightarrow 3^+} g(x) = 2 \\ \lim_{x \rightarrow 1} g(x) = 3 & \lim_{x \rightarrow 3} g(x) = \text{DNE} \end{array}$$

Calculating Limits Using Limit Laws

Theorem (Limit Laws)

Suppose that c is a constant and that the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist ($\pm\infty$ **not allowed**). Then

$$\textcircled{1} \quad \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x).$$

$$\textcircled{2} \quad \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x).$$

$$\textcircled{3} \quad \lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x).$$

$$\textcircled{4} \quad \lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x).$$

$$\textcircled{5} \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0.$$

Here are some other useful limit laws:

- 6 $\lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n$
- 7 $\lim_{x \rightarrow a} c = c.$
- 8 $\lim_{x \rightarrow a} x = a.$
- 9 $\lim_{x \rightarrow a} x^n = a^n.$
- 10 $\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}, \text{ if } a > 0.$
- 11 $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}, \text{ if } \lim_{x \rightarrow a} f(x) > 0.$

Example

Evaluate the limit and justify each step:

$$\begin{aligned} & \lim_{x \rightarrow 5} (2x^2 - 3x + 4) \\ &= \lim_{x \rightarrow 5} (2x^2 - 3x) + \lim_{x \rightarrow 5} 4 && \text{Law} \\ &= \lim_{x \rightarrow 5} (2x^2) - \lim_{x \rightarrow 5} (3x) + \lim_{x \rightarrow 5} 4 && \text{Law} \\ &= 2 \lim_{x \rightarrow 5} x^2 - 3 \lim_{x \rightarrow 5} x + \lim_{x \rightarrow 5} 4 && \text{Law} \\ &= 2 \cdot 5^2 - 3 \cdot 5 + 4 && \text{Laws} \\ &= 39. \end{aligned}$$

Example (Limit Laws)

Evaluate the limit and justify each step:

$$\begin{aligned}
 & \lim_{x \rightarrow 3} \frac{x + 2}{\sqrt{x - 1}(x + 1)^2} \\
 = & \frac{\lim_{x \rightarrow 3} (x + 2)}{\lim_{x \rightarrow 3} (\sqrt{x - 1}(x + 1)^2)} && \text{Law} \\
 = & \frac{\lim_{x \rightarrow 3} (x + 2)}{\lim_{x \rightarrow 3} \sqrt{x - 1} \cdot \lim_{x \rightarrow 3} ((x + 1)^2)} && \text{Law} \\
 = & \frac{\lim_{x \rightarrow 3} (x + 2)}{\sqrt{\lim_{x \rightarrow 3} (x - 1)} (\lim_{x \rightarrow 3} (x + 1))^2} && \text{Laws} \\
 = & \frac{\lim_{x \rightarrow 3} x + \lim_{x \rightarrow 3} 2}{\sqrt{\lim_{x \rightarrow 3} x - \lim_{x \rightarrow 3} 1} (\lim_{x \rightarrow 3} x + \lim_{x \rightarrow 3} 1)^2} && \text{Laws} \\
 = & \frac{3 + 2}{\sqrt{3 - 1} (3 + 1)^2} = \frac{5}{16\sqrt{2}}. && \text{Laws}
 \end{aligned}$$

Recall that every function which can be using the four arithmetic operations $(+, -, *, /)$ and radicals $\sqrt[n]{}$ is an algebraic function.

Theorem (Direct Substitution)

Let f be an algebraic function. Let the point a be in its domain (i.e., $f(a)$ is defined). Then $\lim_{x \rightarrow a} f(x) = f(a)$.

This theorem is a partial case of the following theorem.

Theorem (Can be taken as definition)

*Let f be a **continuous function**. Let the point a be in its domain (i.e., $f(a)$ is defined). Then $\lim_{x \rightarrow a} f(x) = f(a)$.*

Continuous functions will be defined later in this lecture.

Example (Limit with Direct Substitution)

Find $\lim_{x \rightarrow 3} \frac{x + 2}{\sqrt{x - 1}(x + 1)^2}$

Plug in 3: $\frac{(3) + 2}{\sqrt{(3) - 1}((3) + 1)^2} = \frac{5}{16\sqrt{2}}$

Therefore $\lim_{x \rightarrow 3} \frac{x + 2}{\sqrt{x - 1}(x + 1)^2} = \frac{5}{16\sqrt{2}}.$

Example (Limit in Which Direct Substitution Doesn't Work)

Find $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x - 3}{x^2 - 7x + 12}$

Plug in 3: $\frac{(3)^3 - 3(3)^2 + (3) - 3}{(3)^2 - 7(3) + 12} = \frac{0}{0}$

Zero over zero is undefined, so we can't use direct substitution.

When computing a limit as x approaches a , we don't care what happens when $x = a$. This gives the following **useful fact**:

$$\text{If } f(x) = g(x)$$

when $x \neq a$,

$$\text{then } \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x),$$

provided the limit exists.

We can use this fact to find $\lim_{x \rightarrow a} f(x)$ when $f(a)$ has the form $\frac{0}{0}$. In such a case, we use algebra to find a function $g(x)$ that agrees with $f(x)$ at all points except $x = a$. Here are some common techniques.

- 1 Factoring.
- 2 Using a conjugate radical.
- 3 Finding a common denominator.
- 4 **Using Taylor/Maclaurin series expansion. Studied in Calc II.**

Example (Limit with Factoring)

Find $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x - 3}{x^2 - 7x + 12}$

Plug in 3: $\frac{(3)^3 - 3(3)^2 + (3) - 3}{(3)^2 - 7(3) + 12} = -$

Zero over zero is undefined, so we can't use direct substitution.

Factor: $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x - 3}{x^2 - 7x + 12} = \lim_{x \rightarrow 3} \frac{\quad}{\quad}$

$$= \lim_{x \rightarrow 3} \frac{x^2 + 1}{x - 4}$$

Plug in 3: $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x - 3}{x^2 - 7x + 12} = \frac{(3)^2 + 1}{(3) - 4}$

$$= \frac{10}{-1}$$

$$= -10.$$

Example

Find $\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$

Plug in 0: $\frac{\sqrt{(0)^2 + 9} - 3}{(0)^2} = \frac{0}{0}$

Zero over zero is undefined, so we can't use direct substitution.

Multiply top & bottom by (minus) the conjugate radical:

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} &= \lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} \cdot \frac{\sqrt{t^2 + 9} + 3}{\sqrt{t^2 + 9} + 3} \\ &= \lim_{t \rightarrow 0} \frac{(t^2 + 9) - 9}{t^2 (\sqrt{t^2 + 9} + 3)} = \lim_{t \rightarrow 0} \frac{\cancel{t^2}}{\cancel{t^2} (\sqrt{t^2 + 9} + 3)} \\ &= \lim_{t \rightarrow 0} \frac{1}{\sqrt{t^2 + 9} + 3} \end{aligned}$$

Plug in 0: $=$

Example

Find $\lim_{x \rightarrow 1} g(x)$, where

$$g(x) = \begin{cases} x + 1 & \text{if } x \neq 1 \\ \pi & \text{if } x = 1 \end{cases}$$

g agrees with the function $f(x) = x + 1$ at every point except for $x = 1$.

$$\lim_{x \rightarrow 1} g(x) = \lim_{x \rightarrow 1} (x + 1) = 2.$$

Example (Limit with Factoring)

Find $\lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h}$

Plug in 0: $\frac{(3+(0))^2 - 9}{(0)} = \frac{0}{0}$

Zero over zero is undefined, so we can't use direct substitution.

$$\lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h} = \lim_{h \rightarrow 0} \frac{9 + 6h + h^2 - 9}{h} = \lim_{h \rightarrow 0} \frac{6h + h^2}{h}$$

$$\begin{aligned} \text{Factor: } &= \lim_{h \rightarrow 0} \frac{\cancel{h}(6+h)}{\cancel{h}} \\ &= \lim_{h \rightarrow 0} (6+h) \end{aligned}$$

$$\text{Plug in 0: } = (6 + (0)) = 6.$$

Recall:

$$\lim_{x \rightarrow a} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x).$$

We can use this to find the limit of a piecewise defined function, or show that it doesn't exist.

Example

$$f(x) = \begin{cases} \sqrt{x-4} & \text{if } x > 4 \\ 8-2x & \text{if } x < 4 \end{cases}$$

Determine whether $\lim_{x \rightarrow 4} f(x)$ exists.

$$\lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \sqrt{x-4} = \sqrt{4-4} = 0$$

$$\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} (8-2x) = 8-2 \cdot 4 = 0$$

The left and right hand limits are equal. Therefore the limit exists and

$$\lim_{x \rightarrow 4} f(x) = 0.$$

Theorem

If $f(x) \leq g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist as x approaches a , then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x).$$

Theorem (The Squeeze Theorem)

Suppose $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

Then

$$\lim_{x \rightarrow a} g(x) = L.$$

Example

Show that $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{8}{x}\right) = 0$.

WRONG:
$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{8}{x}\right) = \lim_{x \rightarrow 0} x^2 \cdot \lim_{x \rightarrow 0} \sin\left(\frac{8}{x}\right)$$

Doesn't work because $\lim_{x \rightarrow 0} \sin\left(\frac{8}{x}\right)$ doesn't exist.

$$\begin{array}{rclcl} -1 & \leq & \sin\left(\frac{8}{x}\right) & \leq & 1 \\ -x^2 & \leq & x^2 \sin\left(\frac{8}{x}\right) & \leq & x^2 \end{array}$$

$$\lim_{x \rightarrow 0} x^2 = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} (-x^2) = 0.$$

Therefore by the Squeeze Theorem

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{8}{x}\right) = 0.$$